Explainable Artificial Intelligence: Methods, Evaluation and an Empirical Comparison of Post-hoc Explanation Techniques

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Abstract—Today's most accurate learning systems, such as deep networks and boosted tree ensembles, are hard for people to inspect, and this limits how far they can be trusted in medicine, banking and the legal system. Explainable AI (XAI) tries to close that gap by giving humans a view of why a model produced a given output. We survey why XAI is needed, how its methods are grouped, and how the main techniques work: transparent models, LIME, SHAP, gradient-based attribution and counterfactuals. We also discuss how the explanations themselves can be evaluated. To ground the survey, we test five classifiers and four feature-importance methods on the Wisconsin Diagnostic Breast Cancer data. Logistic regression reaches 98.6 % test accuracy, higher than the 95.8 % of gradient boosting. SHAP and a LIME-style surrogate rank features very similarly (Spearman ρ = 0.91), while permutation importance departs from them (ρ = 0.47). Deleting the features that any method ranks highest hurts accuracy much more than deleting random ones. We close with open problems and guidance on choosing an explanation method.

Keywords—explainable AI, interpretability, black-box models, SHAP, LIME, feature attribution, trustworthy machine learning

I. INTRODUCTION

AI systems are now used in decisions that change people's lives, for example reading medical scans, approving loans, shortlisting job applicants and assisting drivers. The strongest models for such tasks, namely deep neural networks, large ensembles and more recently foundation models, contain millions or even billions of parameters, and neither their users nor their builders can read them directly. This is usually called the black-box problem [1]. If a model gives no reasons, a doctor cannot verify a diagnosis, an auditor cannot examine a ruling, and a person affected by the outcome has nothing to contest.

Explainable AI (XAI) is the research area that builds tools for helping people see how such systems arrive at their outputs [1], [2]. Three motives explain its rapid growth. The first is trust and adoption, since users hesitate to rely on something they cannot follow. The second is regulation: the EU General Data Protection Regulation has been read by some as granting a "right to explanation" for automated decisions [3]. The third is debugging, because explanations can expose spurious correlations, inappropriate inputs and unfair behaviour that an accuracy score never shows.

There is also disagreement about how explainability should be obtained. One camp holds that post-hoc explanations of complex models are useful and necessary. Another argues that for high-stakes decisions one should use models that are interpretable from the start rather than explaining opaque ones afterwards [4]. We do not pick a side here; instead we present experimental evidence relevant to both views.

This paper makes four contributions: (i) a taxonomy and overview of the main XAI methods; (ii) a discussion of how explanations can be assessed; (iii) an experiment on a real medical dataset comparing five classifiers and four feature-importance methods; and (iv) practical recommendations and open questions for future work.

II. BACKGROUND AND TAXONOMY

A. Interpretability versus Explainability

The literature frequently uses interpretability and explainability as if they meant the same thing, although they can be separated. Interpretability is the degree to which a person can understand why a decision was made by examining the model directly. Explainability, by contrast, usually refers to additional information or procedures that describe the behaviour of a model that is itself opaque [5], [6]. Lipton separates transparency (for example simulatability, decomposability and algorithmic transparency) from post-hoc interpretation, and cautions that "interpretability" stays vague until the intended meaning is stated [5]. Drawing on social science, Miller observes that human explanations are contrastive, selective and social, and argues that XAI should respect these properties if it is to be useful to people [7].

B. A Taxonomy of XAI Methods

Fig. 1 shows the three axes along which XAI methods are usually organised [8], [2]. Regarding stage, a method is intrinsic when the model is understandable by construction and post-hoc

when the explanation is generated after training. Regarding scope, global explanations describe the behaviour of the model as a whole, whereas local explanations account for one prediction only. Regarding access, model-specific techniques depend on the internal structure of a particular model family, while model-agnostic techniques only query the model as a black box.

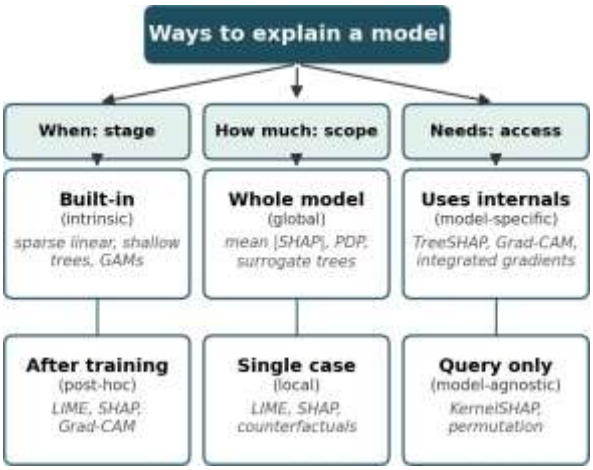


Fig. 1. Three ways of classifying XAI methods (stage, scope and access), with typical examples of each.

C. Desirable Properties of Explanations

A good explanation is expected to be faithful, stable, comprehensible, sparse and actionable. Faithfulness means the explanation matches what the model really does. Stability means that similar inputs receive similar explanations. Comprehensibility means the content suits the knowledge of the person reading it. Sparsity limits the explanation to a handful of factors, and actionability tells the user what could be changed to alter the outcome. These goals often pull against each other: a sparse explanation is quick to read but may leave out part of the model's reasoning, and so be less faithful.

I. Key XAI Techniques

A. Intrinsically Interpretable Models

Linear and logistic regression, small decision trees, rule lists and generalized additive models (GAMs) expose their decision logic directly. GAMs with pairwise interaction terms have been used successfully in healthcare, since the contribution of every feature can be plotted and checked by clinicians [9]. Their limitation is capacity: on images or text they normally fall behind deep networks.

B. LIME: Local Surrogate Models

Local Interpretable Model-agnostic Explanations (LIME) explains a single prediction by slightly perturbing the input and recording how the black-box model f responds to each perturbed copy. A simple, readable model g, usually linear, is then fitted to these responses in the neighbourhood of the instance x [10]. As Eq. (1) shows, the fit minimises a loss L that measures how far g departs from f, where each perturbed sample is weighted by the proximity kernel πx so that nearby points count more. G is the family of interpretable models allowed, and the term $\Omega(g)$ penalises complexity so that g stays simple. LIME works with any model and is easy to use, but its output depends on the random sampling and on the kernel width, which can make explanations unstable [11].

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}}\ \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{1}$$

C. SHAP: Shapley Additive Explanations

SHAP assigns every feature an importance score derived from the Shapley value of cooperative game theory [12].

For a model f, an instance x and the feature set F, the attribution of feature i is a weighted average of its marginal effect over all subsets S of the other features (Eq. 2):

$$\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \left[f_x(S \cup \{i\}) - f_x(S) \right] \tag{2}$$

SHAP yields additive explanations and satisfies the properties of local accuracy, missingness and consistency [13]. Exact computation is expensive because the number of subsets grows exponentially with the number of features, so KernelSHAP approximates the values by sampling, while TreeSHAP computes exact values efficiently for tree ensembles [14]. Averaging the absolute SHAP values over a dataset gives a global importance ranking, so the same framework supplies both local and global explanations.

D. Gradient-Based and Visual Explanations

Gradient-based attribution is the natural choice for neural networks. Grad-CAM weights the feature maps of a convolutional layer by the average gradient of the class score, producing a heat-map of the image regions that mattered for the prediction [15]. Integrated Gradients accumulates gradients along a straight path from a baseline input to the actual input, and satisfies the sensitivity and implementation-invariance axioms [16]. Concept-based approaches such as TCAV go a step further and measure how sensitive the model is to human-understandable concepts instead of raw pixels [17].

E. Counterfactual and Example-Based Explanations

A counterfactual explanation states what would have to change in the input for the model to give a different output. For example, "the loan would have been approved if annual income were higher by a given amount" is a counterfactual. Such explanations can be produced without opening the model [18]. Example-based explanations instead point to training samples that influenced a decision, for instance through influence functions [19]. Both styles resemble how people explain things to each other, and they are useful in finance and

the public sector, where affected individuals need reasons for decisions.

TABLE I
COMPARISON OF REPRESENTATIVE XAI METHODS

Method	Scope	Model access	Strengths	Weaknesses
Linear / tree	Global	Intrinsic	Transparent	Limited capacity
LIME	Local	Agnostic	Simple, flexible	Unstable sampling
SHAP	Both	Agnostic / trees	Axiomatic, consistent	Costly (exact)
Grad-CAM	Local	Gradients	Visual, fast	CNNs only
Int. Gradients	Local	Gradients	Completeness axiom	Baseline choice
Counterfactual	Local	Agnostic	Actionable	May be infeasible

II. EVALUATING EXPLANATIONS

Evaluating an explanation is harder than evaluating a classifier, because there is no ground truth to compare against. Doshi-Velez and Kim describe three levels [6]. Application-grounded evaluation has domain experts judge explanations in the real task. Human-grounded evaluation asks ordinary users to perform simplified tasks. Functionally-grounded evaluation uses formal proxy metrics and needs no human participants. Common functional metrics are fidelity, which measures how closely the explanation reproduces the model's behaviour; stability, which measures how much the explanation changes under small input changes; and deletion or insertion tests, which track how quickly the prediction changes as features are removed or added.

Evaluation matters because an explanation that looks convincing can still be wrong. Adebayo et al. showed that several widely used saliency techniques produce nearly the same maps even when the model has learned nothing, which means such maps do not reflect learning [20]. Other work has shown that some post-hoc methods can be deceived into presenting a discriminatory model as acceptable [21].

III. EMPIRICAL STUDY

A. Dataset and Experimental Setup

We use the Wisconsin Diagnostic Breast Cancer (WBCD) dataset [22], which is bundled with scikit-learn [23]. It holds 569 samples, of which 357 are benign and 212 malignant, each described by 30 numeric features. The features are the mean, the standard error and the "worst" value of ten nuclear measurements (such as radius, texture, area and concavity) computed from digitised images of fine-needle aspirates of breast masses. A 75/25 split gave 426 training and 143 test samples.

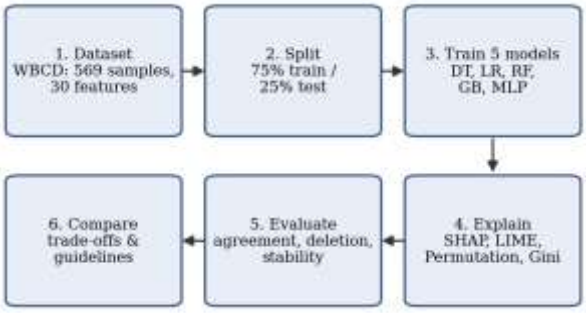


Fig. 2. Experimental workflow: dataset, split, model training, explanation generation, evaluation and comparison.

B. Models and Feature-Importance Methods

Five classifiers were trained: a depth-3 decision tree (DT), logistic regression on standardised inputs (LR), a random forest of 300 trees (RF) [24], a gradient-boosting classifier (GB) with default settings [25], and a multilayer perceptron (MLP) with two hidden layers of 64 and 32 units. For the GB model we computed four global feature-importance rankings: (a) mean absolute SHAP value from TreeSHAP; (b) permutation importance on the test set with 30 repeats; (c) impurity-based Gini importance; and (d) a LIME-style score obtained by fitting a locally weighted ridge surrogate to 1,500 Gaussian perturbations around each of 40 samples. Because the LIME-style score is our own implementation of the idea, its values may differ from those of the reference package.

C. Accuracy versus Interpretability

Table II and Fig. 3 report the results. Logistic regression, an inherently interpretable model, reached 98.6 % test accuracy and 98.1 % cross-validated accuracy, ahead of every other model, including the black-box GB (95.8 % and 94.6 %), the random forest (95.8 % and 96.0 %) and the MLP (97.2 % and 97.7 %). The shallow tree, the most interpretable of the five, scored lowest at 94.4 %. Since the test set has 143 samples, one extra error lowers accuracy by about 0.7 points, so small gaps should be read with care. Even so, the results indicate that on small, well-curated tabular data the trade-off between interpretability and accuracy is not unavoidable.

TABLE II
CLASSIFICATION PERFORMANCE ON WBCD (%)

Model	Test acc.	Test AUC	5-fold CV acc.
Decision tree (d=3)	94.4	94.63	92.5 ± 2.8
Logistic regression	98.6	99.77	98.1 ± 0.6
Random forest	95.8	99.47	96.0 ± 1.8
Gradient boosting	95.8	99.20	94.6 ± 3.8
MLP (64–32)	97.2	99.58	97.7 ± 1.1

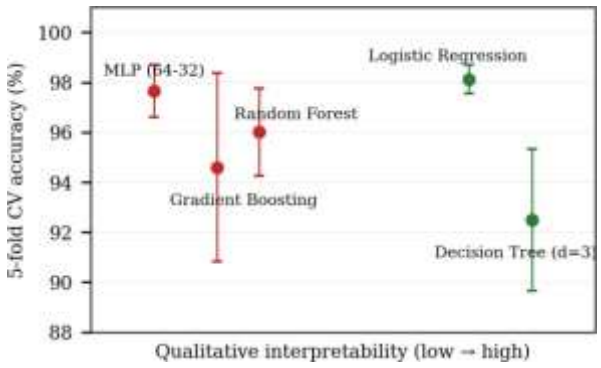


Fig. 3. Five-fold CV accuracy (mean ± s.d.) against a qualitative interpretability ordering; the interpretability positions are illustrative and were not measured.

D. Global and Local Explanations

Fig. 4 shows the global SHAP importance of the GB model. The top features are worst concave points, worst area, worst radius, worst perimeter and area error, all of which describe the size and irregularity of the cell nuclei. This agrees with clinical understanding that malignant nuclei tend to be larger and more irregular, so the model appears to rely on sensible signals. Fig. 5 gives the local explanation for one test sample, showing how the SHAP value of each feature moves the predicted probability of benign, which is 0.9994 for this sample.

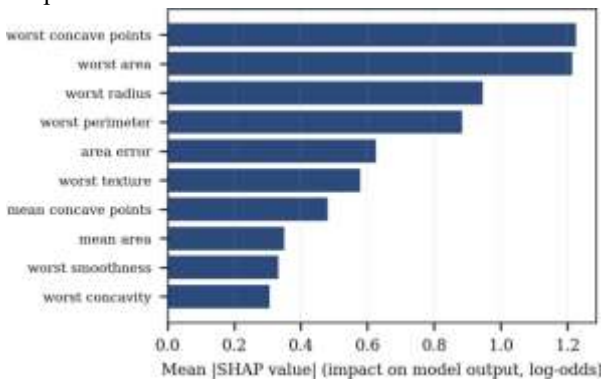


Fig. 4. Global importance for the gradient-boosting model: the ten largest mean |SHAP| values on the test set.

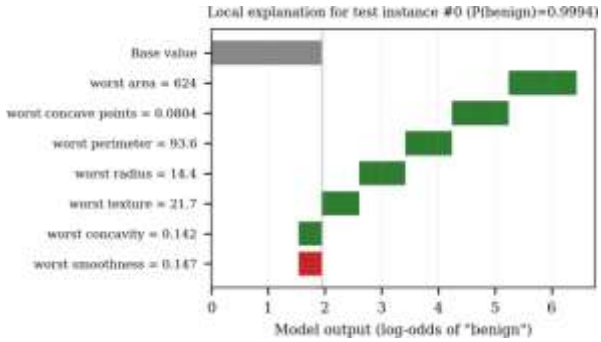


Fig. 5. Local SHAP explanation for one test sample. Green bars push the prediction toward benign and red bars toward malignant.

E. Agreement and Stability of Explanation Methods

Table III lists the Spearman rank correlations between the four importance rankings. SHAP and the LIME-style ranking agree strongly ($\rho = 0.91$), and both correlate reasonably with Gini importance. Permutation importance is the outlier ($\rho = 0.47$ with SHAP). A likely reason is the group of highly correlated features such as radius, perimeter and area: when one of them is permuted, the model can fall back on the others, so the importance is shared among them or understated. SHAP, by comparison, spreads credit across such features.

TABLE III
SPEARMAN RANK CORRELATION BETWEEN IMPORTANCE RANKINGS

	SHAP	**LIME-style**	**Permutation**	**Gini**
SHAP	1.00	0.91	0.47	0.73
LIME-style	0.91	1.00	0.52	0.79
Permutation	0.47	0.52	1.00	0.34
Gini	0.73	0.79	0.34	1.00

We also checked stability. The LIME-style explanation of one fixed sample was generated in ten runs with different random seeds, and the mean rank correlation between the resulting coefficient magnitudes was only 0.70. TreeSHAP uses no randomness and returns identical output every time. This confirms the sampling variability of LIME-type methods reported in [11].

F. Deletion Test

To see how much the model depends on the ranked features, we replaced the top-k features of each ranking with their training-set mean and measured the change in test accuracy of the GB model (Fig. 6, Table IV). Replacing 10 random features lowered accuracy only slightly, from 95.8 % to 94.8 %. Replacing the 10 top features by SHAP, permutation importance or Gini lowered it sharply, to 79.0 %, 74.1 % and 76.2 % respectively, and the LIME-style ranking gave 86.7 %. So all four methods point to features the model really uses. SHAP has an effect early on ($k = 5$), whereas permutation importance gives the lowest mean accuracy over $k = 1$ to 15 (Table IV). This is expected, because permutation importance is computed on the test set itself, which favours it in this test. The curves are not always monotonic: when features are set to their means, some samples land in regions the model never saw in training, which can produce irregular changes.

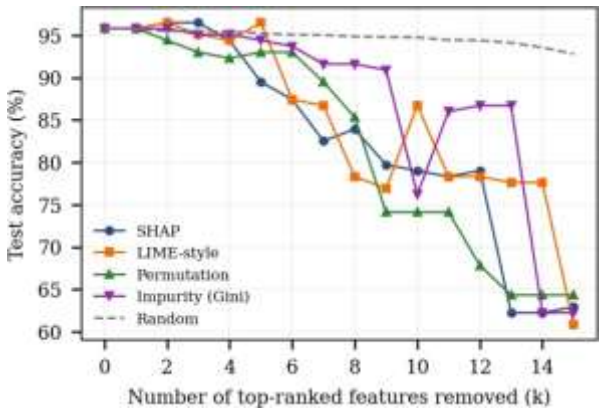


Fig. 6. Deletion test: test accuracy of the gradient-boosting model after the top-k features of each ranking are replaced by their means. A faster drop indicates a better ranking.

TABLE IV
TEST ACCURACY (%) AFTER REMOVING TOP-K FEATURES

Ranking	k = 5	k = 10	Mean (k = 1–15)
SHAP	89.5	79.0	82.0
LIME-style	96.5	86.7	84.5
Permutation	93.0	74.1	81.3
Gini	94.4	76.2	86.9
Random (avg. of 20)	95.2	94.8	94.8

G. Discussion

Three observations stand out. First, accuracy did not follow model complexity: the linear model finished ahead of the ensembles and the neural network, which suggests that the WBCD features may be close to linearly separable and that extra capacity adds little here. Second, the rankings differ mainly in how they treat the correlated size features. SHAP and the LIME-style score reward the group as a whole, whereas permutation importance can give each member a low score because the others compensate, so a practitioner who reads only one ranking may form a different picture of the model than one who reads two. Third, agreement between methods was not the same as faithfulness: the LIME-style ranking correlated strongly with SHAP ($\rho = 0.91$) yet was the least damaging of the four rankings in the deletion test at k = 5 and k = 10 (96.5 % and 86.7 %). A high correlation therefore shows that two explanations resemble each other, not that either one is correct, which is why we pair an agreement check with a deletion or sanity test.

H. Limitations

This study uses a single tabular dataset, one train/test split and one model for the explanation experiments, and deletion is only one of many ways to measure faithfulness. No human evaluation was carried out. The results should therefore be read as an illustration and not as a benchmark.

IV. APPLICATIONS

Healthcare. Explanations let a clinician check whether a diagnostic model attends to clinically meaningful evidence, such as nuclear shape in tabular features or the lesion area highlighted by Grad-CAM in medical images [15]. *Finance.* Credit-scoring and fraud models must justify their decisions, especially rejections, to customers and auditors, and counterfactuals offer a practical way to do so [18]. *Safety-critical systems.* Explanations help engineers find which sensor input caused a particular system action. *Law and public policy.* Algorithmic decisions in justice and welfare must be open to contestation [3], [4].

V. CHALLENGES AND FUTURE DIRECTIONS

- **Faithfulness versus plausibility.** An explanation that looks plausible to a human need not match what the model computes [20], so explanations should be validated quantitatively.
- **Correlated features and causality.** Attribution methods describe associations inside the model, not causal effects in the world, and correlated inputs can split or hide importance, as Table III shows.
- **Robustness and security.** Explanations can be fragile and can be manipulated on purpose [11], [21].
- **Human-centred evaluation.** Few methods have been tested in user studies, and standard protocols for such tests are still needed [6], [7].
- **Large language and foundation models.** The rationales produced by large language models can read fluently without reflecting the model's real reasoning [26].

VI. CONCLUSION

Interpretability matters before machine learning is placed in high-stakes settings. This paper surveyed the main ideas and methods of XAI and then tested them in an experiment. On the WBCD data, an interpretable logistic regression model matched or beat the black-box models, SHAP gave deterministic explanations that were consistent at both global and local level and agreed with domain knowledge, and the four importance rankings were only partly similar even though each picked out genuinely important features. Based on this, we recommend (i) trying interpretable models first, (ii) using at least two XAI tools and checking that they agree, (iii) validating explanations with deletion tests or sanity checks, and (iv) involving domain experts in the validation. Future work will extend the study to image and text models and add human-subject experiments.

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